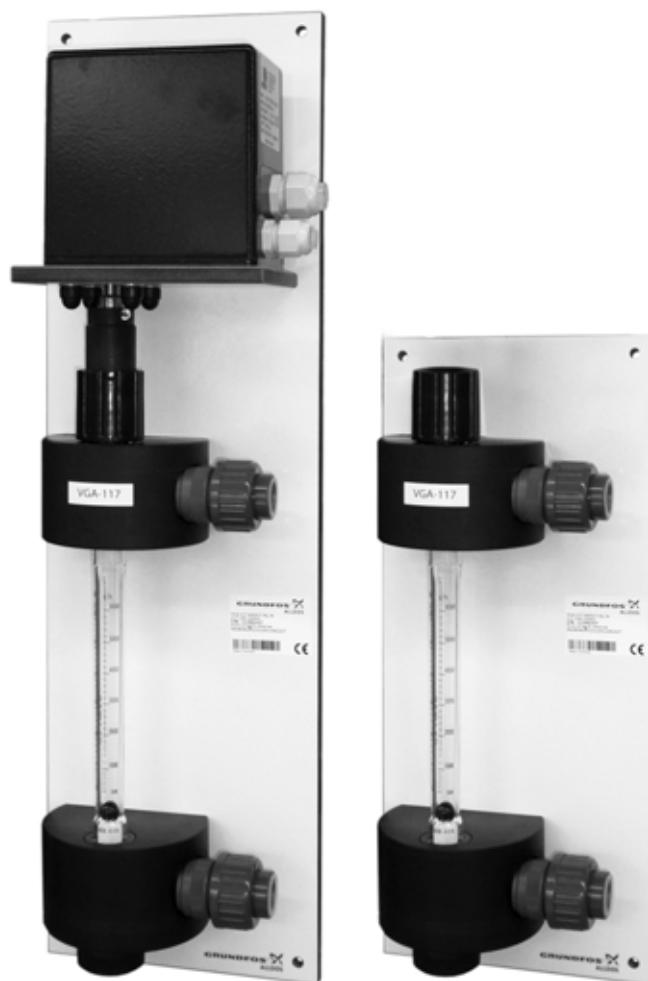


Vaccuperm VGA-117

Gas dosing regulator

Installation and operating instructions



Other languages

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Original installation and operating instructions.

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**Warning**

Prior to installation, read these installation and operating instructions. Installation and operation must comply with local regulations and accepted codes of good practice.

1. Safety instructions

These installation and operating instructions contain important information for the user of the product:

- Technical data
- Instructions for commissioning and use
- Safety instructions

If you require further information, or if problems occur that are not described in detail in this manual, please contact Grundfos.

1.1 Symbols used in this document

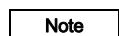
This manual contains the following standardised safety instructions about possible residual risks:

**Warning**

If these safety instructions are not observed, it may result in personal injury.

**Caution**

If these safety instructions are not observed, it may result in malfunction or damage to the equipment.

**Note**

Notes or instructions that make the job easier and ensure safe operation.

1.2 Safety advice for handling chlorine**1.2.1 Hazards of chlorine gas****Warning**

Toxic by inhalation.

Irritating to eyes, respiratory system and skin.

- Causes whooping cough, dyspnoea and lacrimation.
- Has a slight paralysing effect to the central nervous system.
- Concentrations of more than 10 ppm chlorine gas in the respiratory air mean an acute danger to life.
- Inhaling air with a high concentration of chlorine gas for a long time is lethal.

1.2.2 Hazards of liquid chlorine

- Causes burns of skin.
- Causes reddening and blistering of skin.

1.2.3 Rules of conduct

- Change chlorine containers only with gas mask.
- Enter contaminated rooms only with protective suit and compressed-air respirator.
- In case of flight wear gas mask, if possible. Observe wind direction.

1.3 Protective equipment

1.3.1 Personal protective equipment

The operating authority of a chlorine gas dosing system has to provide respiratory equipment (full-sight gas mask), personally fitted, labelled by name, with an effective chlorine filter (B2P3) and at least 1 spare filter per gas mask, for each operator. The protective equipment must be stored easily accessible outside the chlorine rooms.

- For systems with chlorine drums at least 2 protective suits with compressed-air respirators must be available.

1.3.2 Obligations of the operating authority

- Introducing the operators to handling the protective equipment.
- The operators must use the personal protective equipment regularly, or carry out exercises with it at least every 6 months.
- The gas mask filters must be replaced regularly
 - after the expiry of the date of durability
 - at least 6 months after opening (note the opening date on the filter)
 - after contact with chlorine.
- Observe employing prohibition according to the local laws.

1.3.3 Storage of the protective equipment

- Outside the chlorine rooms
- Well visible
- Easily available at any time
- Protected from dust and moisture.

1.4 First aid in case of accidents

1.4.1 First aid after inhaling chlorine

1. Keep calm.
2. Remove injured persons from the dangerous area.
3. Helpers must pay attention to personal protection.
4. Remove contaminated clothes.
5. Calm down injured persons and keep them warm with blankets.
6. Supply fresh air, use oxygen respirator, if possible.
 - No mouth-to-mouth resuscitation!
7. Call medical aid or transport to hospital
 - lying
 - sitting in case of difficulty in breathing.
 - State chlorine as the cause.

1.4.2 First aid after chemical burn of skin

1. Keep calm.
2. Remove contaminated clothes.
3. Rinse skin with plenty of water.
4. Bandage the wound aseptically.
5. Seek medical aid.
 - State chlorine as the cause.

1.4.3 First aid after chemical burn of eyes

1. Keep calm.
2. Rinse eyes with plenty of water while the patient is lying.
 - Protect healthy eye, if necessary.
 - Spread eyelids widely, let the eye move to all sides.
3. Seek ophthalmologist.
 - State chlorine as the cause.

1.4.4 First aid after internal chemical burn

1. Keep calm.
2. Drink water in short sips.
 - If possible, take medical charcoal.
3. Seek medical aid.
 - State chlorine as the cause.

1.5 Handling chlorine containers

Chlorine is stored in grey steel cylinders or drums in lockable chlorine rooms. Due to safety precautions, chlorine containers are only filled up to 95 % of their capacity.

Kind of gas, weight, owner, producing date and date of the last testing have to be noted clearly on the container.

Observe the following:

- Store chlorine containers fireproof.
- Protect chlorine containers from heat and sunlight.
- Chlorine containers should not be modified or repaired by the user.
- Keep full and empty containers closed.

1.5.1 Steel cylinders

- Contents: 50 kg or 65 kg
- Extract chlorine gas from the upright standing cylinder via the valve.

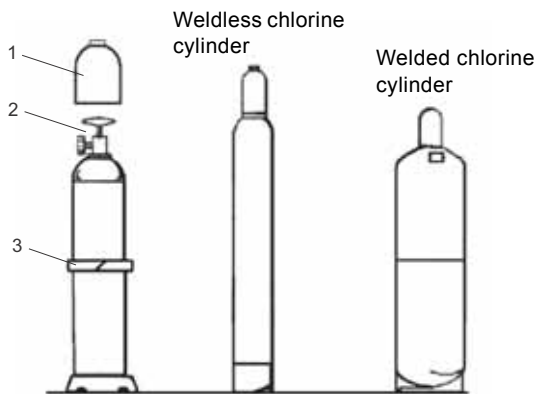


Fig. 1 Design of cylinder

Pos.	Description
1	Protective cap
2	Valve
3	Holding clip for securing the cylinder

1.5.2 Chlorine drums

- Contents: 500 kg or 1000 kg
- Extraction of chlorine gas via the riser pipe valve.
- Extraction of liquid chlorine via the dip pipe valve.

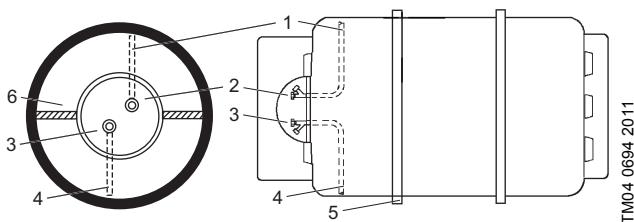


Fig. 2 Design of drum

Pos.	Description
1	Riser pipe for chlorine gas extraction
2	Valve for chlorine gas extraction
3	Valve for liquid chlorine extraction
4	Dip pipe for liquid chlorine extraction
5	Tyre
6	Mark for adjustment of drum

1.5.3 Basic rules



Warning

Handling of chlorine containers only by experienced personnel.

Observe safety regulations for chlorine containers.

- Treat containers carefully, do not throw!
- Protect containers from turning over or rolling away, e.g. with chains or clips.
- Protect containers from direct sunlight and temperatures over 50 °C.
- Transport containers only with valve protection nut and protective cap.



Warning

These rules apply for both full and empty containers, as empty containers still contain rests of chlorine and therefore are under pressure.

1.5.4 Valid regulations



Warning

Local laws and regulations for handling, transport and storage of chlorine must be observed.

In Germany, the following regulations apply:

- Regulations for accident prevention "Chlorination of water" with process instructions (GUV-V D5)
- Regulations concerning places of work (ArbStättV)
- Technical rules for pressurized gases (TRG 280, 310 and 330)
- Operation of baths (BGR 108)
- Chlorine gas dosing systems for water treatment (DIN 19606)
- Chlorine for water treatment (DIN 19607)

1.6 Chlorine rooms

Chlorine rooms are rooms, where a chlorine gas dosing system and/or chlorine containers are located. The chlorine in these rooms is under pressure.

1.6.1 Regulations for chlorine rooms

According to the German regulations for accident prevention "Chlorination of water", such rooms have to meet the following requirements:

- They should not have connections to other rooms, and must be separated gas-tight and fire-resistant.
- They must be at ground level, dry and air-conditioned. The recommended room temperature is 18-20 °C. It should be neither lower than 0 °C nor higher than 50 °C.
- Overpressure lines of dosing equipment must not end in the open air.
- Ventilation openings leading into the open air must be limited to 2 x 20 cm².
- Appropriate exhaust devices with downstream absorption systems must be installed.
- Chlorine gas must not get into lower-lying rooms, shafts, pits, canals or aspirating holes of ventilation systems.
- Chlorine rooms must be equipped with a water sprinkling system to precipitate the escaping chlorine gas.
 - The sprinkling system must have a run-off with air trap.
 - Operation must be possible manually from outside the chlorine rooms.
- A chlorine gas warning system must be installed
 - with optical and acoustical alarm device,
 - with connection to the water sprinkling system,
 - with a warning system that reactivates automatically after switch-off (e.g. for container exchange).
- The floor of a chlorine room must be even, the exit not below ground level and not higher than a possible loading ramp.

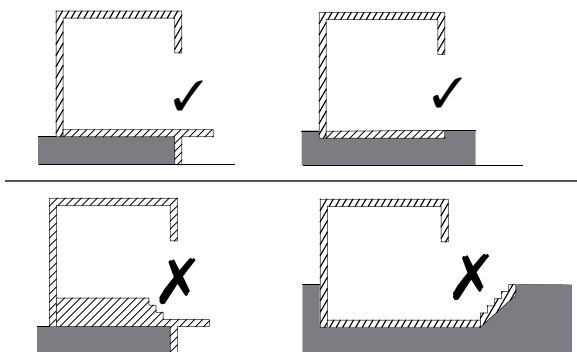


Fig. 3 Requirements of chlorine rooms

- Chlorine rooms must have a direct exit to the open air.
- Doors must be lockable, open outwards, and it must be possible to open the doors without a key from inside the room.
- Chlorine rooms must not be dedicated for the permanent stay of people.
- Eating, drinking, or storing food in chlorine rooms is prohibited.

1.6.2 Labelling of chlorine rooms

In Germany, according to DIN 4844, the following warning signs must be installed outside the entrance of a chlorine room:



Fig. 4 Warning sign and supplementary sign "Chlorination plant: Access only for instructed persons"

Inside a chlorine room, the mandatory sign "Use the gas mask" and the "Instruction sheet for first aid in case of chlorine gas intoxications" must be installed.



Fig. 5 Mandatory sign (DIN 4844)

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2. Introduction

2.1 Correct usage

The VGA-117 dosing regulator is exclusively designed for dosing chlorine gas into a vacuum line.

2.2 Inappropriate usage

Operational safety is only guaranteed, if the product is used correctly. All operating methods conflicting with correct usage are not permitted, and lead to the expiry of all liability claims.



Warning

Unauthorised structural modifications to the product may result in serious damage to equipment and personal injury.

It is forbidden to open, modify, bridge, remove, bypass or disable components, especially safety equipment.

2.3 Qualification of the users

Based on his professional training and experience, the user must have a good knowledge of the operation of chlorine gas dosing systems. He must be so familiar with the relevant laws and regulations governing safety in the workplace and accident prevention, and with the recognised standards of good practice, that he is able to judge the safe working condition of a chlorine gas dosing system.

The user is responsible for operation and monitoring of the product at the installation site.

The responsibilities of the user:

- Read this manual before operating.
- Be trained by qualified Grundfos personnel in the operation of the product.
- Observe the relevant regulations governing safety in the workplace and accident prevention.
- Wear appropriate protective clothing in accordance with national regulations for the prevention of accidents when operating the system and handling chemicals.

2.4 Responsibilities of the operator

The operator of the product is responsible for the following:

- Ensure that this manual is kept clearly accessible in the immediate vicinity of the system for the entire service life.
- Meet the installation requirements specified by the manufacturer (environmental conditions, required water connections and fittings, electrical connection, protective pipe for dosing line if necessary, audible or optical warning device if necessary).
- Ensure that water lines and armatures are regularly checked, serviced and maintained.
- Obtain official approval for storing chemicals, if necessary.
- Train users in the operation of the system.
- Ensure that the regulations for the prevention of accidents are observed in the installation site.
- Provide all users and service personnel with protective clothing (e.g. face mask, gloves, protective apron) in accordance with the national regulations.

2.5 Maintenance and service personnel

The product may only be maintained and serviced by personnel authorised by Grundfos.

2.6 Type key VGA-117 dosing regulator

The type key serves for the identification of the product, not for configuration purposes.

Code	Example	VGA-117	-1000	/5	-M	O	, N
VGA	Vaccuperm Gas Advanced						
	Dosing flow						
500	25-500 g/h						
1000	50-1,000 g/h						
2000	100-2,000 g/h						
4000	200-4,000 g/h						
6000	300-6,000 g/h						
8000	400-8,000 g/h						
10000	500-10,000 g/h						
	Connection input and output						
1	PE hose 8/11						
2	PE hose 10/14						
3	PVC pipe DN15 (d 20)						
	Adjustment of the dosing flow						
M	Manual						
A	Automatic, 230-240 V, 50/60 Hz						
B	Automatic, 110-115 V, 50/60 Hz						
	Control						
0	Without						
1	4-20 mA						
2	Direct						
	Differential pressure regulator						
Y	Yes, with vacuummeter						
N	No						

2.7 Transport and storage

- Transport carefully, do not throw!
- Store in a dry and cool place.

2.8 Unpacking

Observe when unpacking:

- The delivery must be complete.
- No humidity and no foreign matters should get into gas-leading parts.
- Mount the components of the dosing system as soon as possible after unpacking.

3. Technical data

3.1 General technical data

Accuracy	± 4 % of upper limit
Permissible medium	Cl ₂
Setting range	1:20
Permissible vacuum regulator	VGA-111, VGA-146 (up to 4000 g/h)
Flowmeter	According to the floater principle, ball in measuring glass, length of measuring glass 190 mm
Weight	Without servomotor: 2.6 kg With servomotor: 4.2 kg
Vacuum line between dosing regulator and injector	PE hose 8/11 or 10/14 PVC tube DN 15 (d20)

3.4 Dimensions

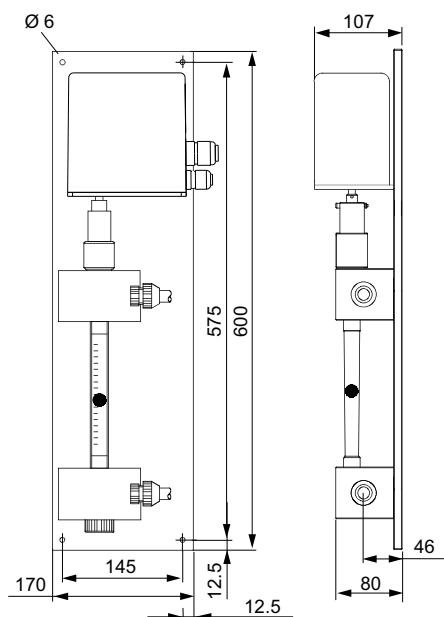


Fig. 6 VGA-117 dosing regulator with and without servomotor

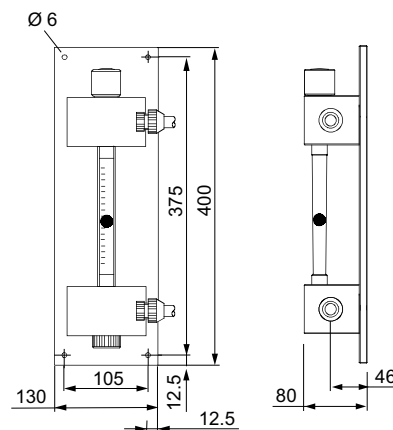
3.2 Dosing flow

50-1,000 g/h	2-50 lbs/day
100-2,000 g/h	5-100 lbs/day
200-4,000 g/h	10-210 lbs/day
300-6,000 g/h	15-310 lbs/day
400-8,000 g/h	20-420 lbs/day
500-10,000 g/h	30-520 lbs/day

3.3 Accessories

The following accessories are not included in the standard delivery:

- Hoses of different lengths
- Holding plate for wall fixing when changing containers
- Ammonia water for tightness check.



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3.5 Electrical data of servomotor

3.5.1 Servomotor with reversing potentiometer

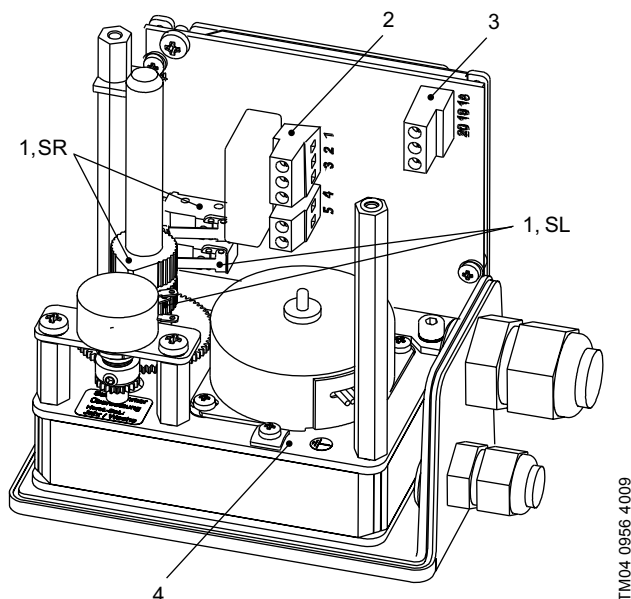


Fig. 7 Servomotor with reversing potentiometer

Pos.	Description
1	SR, limit switch cam min. SL, limit switch cam max.
2	Supply voltage input
3	Potentiometer input/output
4	Earth terminal

Mains voltage	220-240 V $\pm$ 10 %, 50/60 Hz $\pm$ 5 %, 100 % ED (Standard) 110-120 V $\pm$ 10 %, 50/60 Hz $\pm$ 5 %, 24 V AC $\pm$ 10 %, 50/60 Hz $\pm$ 5 %
Rated current	30 mA (at 230 V), 60 mA (at 115 V), 240 mA (at 24 V)
Load capacity of limit switch	2 A at 250 V
Power consumption	Approx. 2 VA
Control variant	Reversing potentiometer 1 k Ω Open/Close signal
Actuating time	90 s for 270 ° at 50 Hz 75 s for 270 ° at 60 Hz
Adjustment angle	Max. 270 °
Protection class	IP65
Permissible ambient temperature	-15 °C to 60 °C

3.5.2 Servomotor with analog control (4-20 mA/0-20 mA)

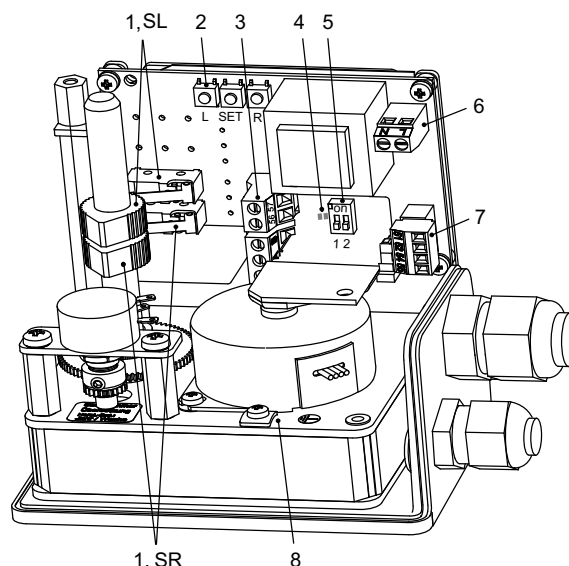


Fig. 8 Servomotor with analog control

Pos.	Description
1	SR, limit switch cam min. SL, limit switch cam max.
2	Programming buttons
3	Alarm output
4	LEDs
5	DIP switch
6	Mains connection terminals
7	Analog signal connection terminals
8	Earth terminal

Mains voltage	220-240 V $\pm$ 10 %, 50/60 Hz $\pm$ 5 %, 100 % ED (standard) 110-120 V $\pm$ 10 %, 50/60 Hz $\pm$ 5 %, 24 V AC $\pm$ 10 %, 50/60 Hz $\pm$ 5 %
Rated current	30 mA (at 230 V), 60 mA (at 115 V), 240 mA (at 24 V)
Load capacity of limit switch	2 A at 250 V
Power consumption	Approx. 2 VA
Control variant	Analog control Input/output
Signal input, signal output	4-20 mA (standard setting) 0-20 mA (switchable)
Setpoint input	4 (0)-20 mA (DC) Load 250 Ω Overload protection 25 mA Protection against reverse polarity to -25 mA ESD protection, input filter Resolution: 10 bit
Actual value output	4 (0)-20 mA Load max. 500 Ω Resolution: 10 bit
Carrying capacity of signal output	Output current: 4-20 mA Minimum voltage for open circuit 10 V
Potentiometer	1000 Ω
Actuating time	90 s for 270 ° at 50 Hz 75 s for 270 ° at 60 Hz
Adjustment angle	Max. 270 °
Protection class	IP65
Permissible ambient temperature	0 °C to 60 °C

3.6 Physical and chemical data of chlorine

Under normal conditions of pressure and temperature, chlorine is a yellowish green gas with a pungent odour, which exists as Cl₂ molecule.

It is not flammable, but can promote the flammability of metals, hydrocarbons, etc.

Atomic weight	35.457 u
Molecular weight Cl ₂	70.941 u
Density (liquid)	1.57 g/cm ³ at -34.05 °C
Density (gas)	3.214 g/l at 0 °C, 1 bar
1 litre of liquid chlorine at 0 °C	is equivalent to 457 litres (0.457 m ³) of chlorine gas
1 kg of liquid chlorine at 0 °C	is equivalent to 311 litres (0.311 m ³) of chlorine gas
Specific gravity	2.486 (specific gravity of air: 1)
Boiling point	-34.05 °C (1 bar)
Melting point	-101.5 °C
Evaporation heat	269 kJ/kg (at 0 °C)
Heat conductivity	0.527 kJ/m ² h (liquid chlorine)
Degree of purity according to DIN 19607	99.5 %
TLV (Threshold Limit Value)	1.5 mg/m ³ (0.5 ppmv)

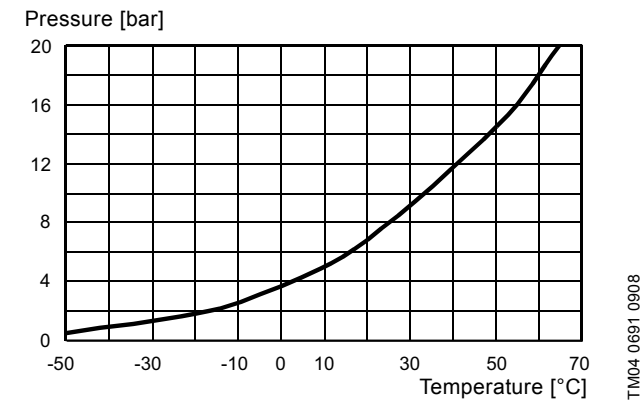


Fig. 9 Vapour pressure curve of chlorine

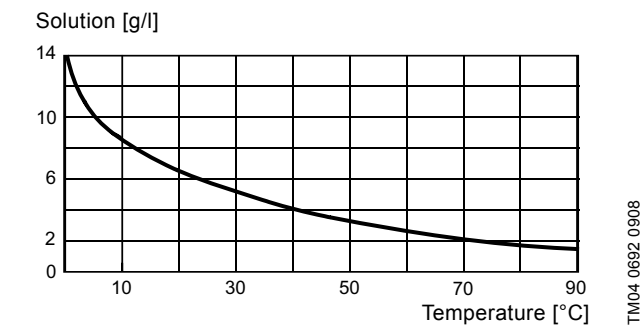


Fig. 10 Solubility of chlorine gas in water

4. Design and function

4.1 Description of the device

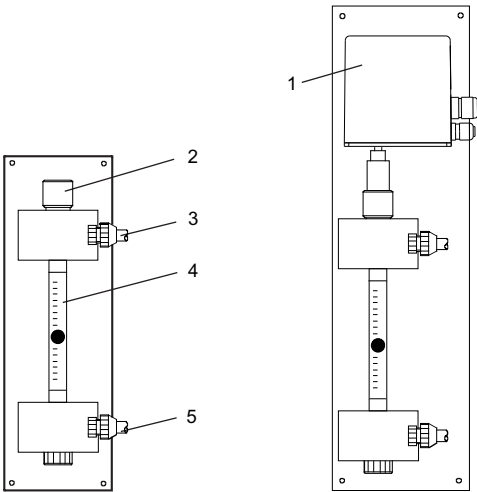


Fig. 11 Dosing regulator without and with servomotor

Pos.	Description
1	Servomotor (option)
2	Adjustment knob for setting the dosing flow
3	Connection for the vacuum line (to the injector)
4	Measuring glass for indication of the dosing flow
5	Connection for the vacuum line (from the vacuum regulator)

Caution

The rate valve is not a perfectly closing shut-off valve. Do not try to stop the chlorine gas dosing system by closing the rate valve!
Switching off the chlorine gas dosing system, see section 7.5.3 *Switching off while the system is running*.

4.2 Functional principle

4.2.1 Dosing regulator

The dosing regulator regulates the chlorine gas flow with a valve.
The flow is read at the top of the ball in the measuring glass.

4.2.2 Vacuum chlorine gas dosing system

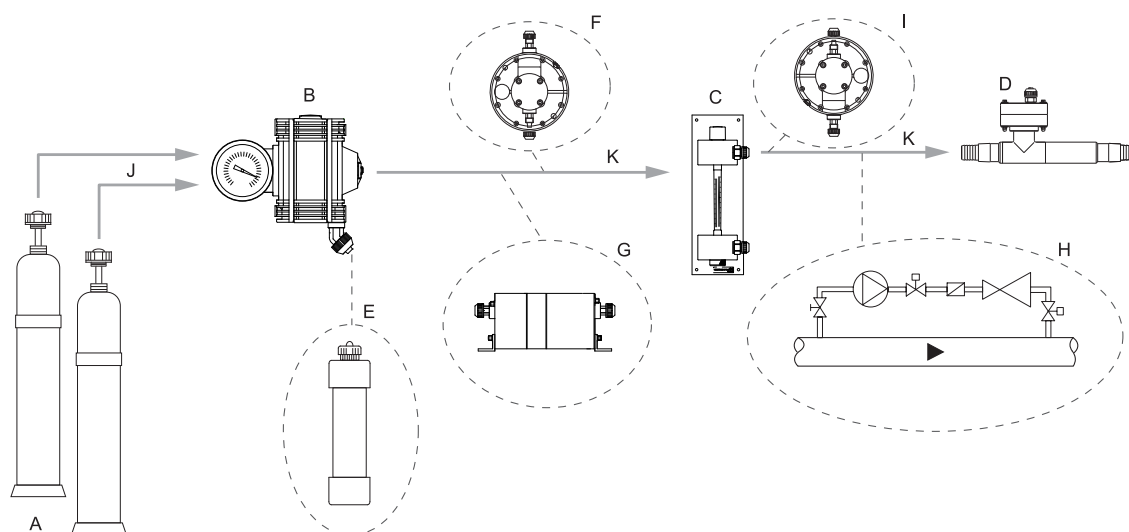


Fig. 12 Principle of a vacuum chlorine gas dosing system

Pos.	Description
A	Chlorine cylinder
B	Vacuum regulator
C	Dosing regulator
D	Injector
E	Adsorption filter (option)
F	Safety valve (option)
G	Change-over device (option)
H	Water apparatus (option)
I	Differential pressure regulator (option)
J	Chlorine gas pressure line
K	Chlorine gas vacuum line

Functioning of a vacuum chlorine gas dosing system

Handling, transport and storage of chlorine for the disinfection of drinking and pool water are a challenge for plant engineers. This is the reason why the vacuum principle has been used in dosing systems for a long time. The pressure of the chlorine gas is reduced to a vacuum.

This method prevents escape of chlorine gas. In the event of a pipe burst no chlorine gas can escape, only ambient air is drawn in.

Vacuum chlorine gas dosing systems are composed of three principal components.

Vacuum regulator (B)

The vacuum regulator is a pressure reducing valve, which reduces the overpressure from the chlorine tank side to the negative pressure on the vacuum side. The valve opens, when a sufficient vacuum has built up on the outlet side. Vacuum regulators with pressure gauge and liquid trap are available for more safety.

Dosing regulator (C)

The chlorine gas flow is adjusted with the dosing regulator. This can be effected manually or automatically via motor control.

Injector (D)

Injectors have the task to bring chlorine gas into the water flow. They operate according to the principle of water jet pumps. The injector body comprises a nozzle with a successional diffuser.

Between the nozzle and the diffuser there is a narrow annular gap, where chlorine gas is sucked out of the dosing line through the injector head. A diaphragm non-return valve at the end of the vacuum line prevents the ingress of water into the vacuum line.

Differential pressure regulator (I)

Regulates the difference of the pressures before and after the rate valve to a constant value. Adjusted dosing flow stays constant even when the injector vacuum varies.

5. Assembly and installation

5.1 Selection of vacuum lines



Warning
Observe section [1. Safety instructions](#).

The vacuum needed for the transport of chlorine gas is built up by the injector, and maintained by the vacuum lines. Rigid PVC pipes or flexible PE hoses are used as vacuum lines.

The following tables show the recommended diameter of vacuum lines, depending on the line length and dosing quantity.

5.1.1 Vacuum lines between vacuum regulator and dosing regulator

Max. length [m] of the vacuum lines between vacuum regulator and dosing regulator*	Dosing quantity [g/h]						
	40	100	250	500	1000	2000	4000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 15	DN 15
50	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15
75	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 15
100	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15	DN 20

* For the calculation, a pressure loss of $p = 12.5$ mbar in the vacuum line was assumed.

5.1.2 Vacuum lines between dosing regulator and injector

Max. length [m] of the vacuum lines between dosing regulator and injector**	Dosing quantity [g/h]						
	40	100	250	500	1000	2000	4000
0	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8
10	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8
20	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
30	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
40	DN 8	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10
50	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
75	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15
100	DN 8	DN 8	DN 8	DN 8	DN 8	DN 10	DN 15

** For the calculation, a pressure loss of $p = 50$ mbar in the vacuum line was assumed.

5.2 Connecting the dosing regulator



Warning

Before connecting, make sure that the valves of all chlorine containers are closed.

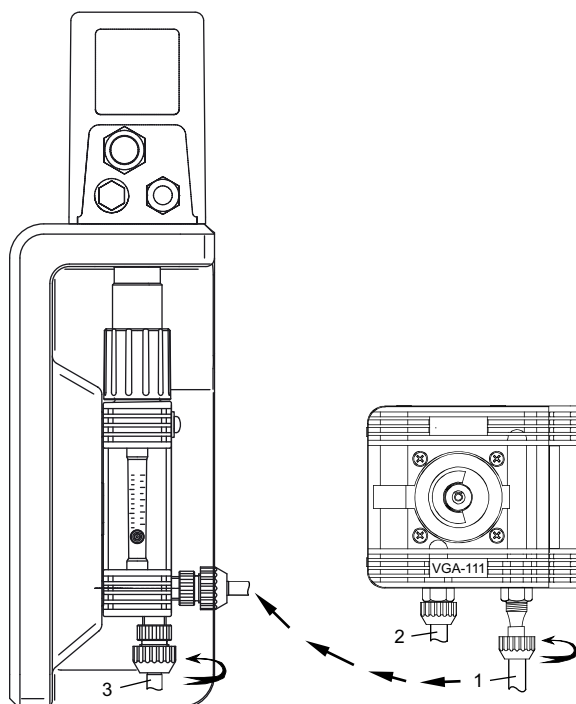


Fig. 13 Vacuum connections

Pos.	Description
1	Vacuum line (to the injector)
2	Vacuum line (to the vacuum regulator)
3	Servomotor
4	Vacuummeter
5	Differential pressure regulator

For details on vacuum lines, see section [5.1 Selection of vacuum lines](#)

Caution

Only tighten the union nuts of the vacuum connections by hand. Do not use any tools. Danger of damages!

Make sure that the vacuum connections are clean and dry.

The dosing regulator can be installed outside the chlorine room.

1. Connect the dosing regulator to the line from the vacuum regulator (2).
2. Connect the line (1) from the dosing regulator, or from the optional differential pressure regulator, to the injector.

5.3 Connecting an external controller to a servomotor with analog control

5.3.1 Warnings and notes for electrical connection

Warning

Disconnect servomotor from mains before removing the cover.

Observe the local safety regulations.

Protect unused screwings, cable connections and plugs against corrosion and moisture with suitable caps.

To ensure electromagnetic compatibility (EMC) signal lines must be shielded, and guided in different cable channels than mains cables.

Caution

The shield must be connected to PE or ground on the side of the counterpart.

Before connecting the mains cable check, if the mains voltage indicated on the type plate corresponds to the local mains voltage. Wrong voltage can lead to damages of the servomotor.

Caution

Depending on the optional components selected, connect mains plug to mains directly or via the terminal box.

Note

5.3.2 Circuit diagram servomotor with analog control

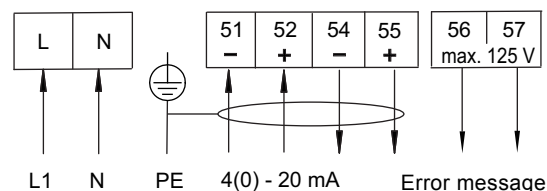


Fig. 14 Servomotor with analog control

5.3.3 Connecting the external controller Conex® DIA-2Q

Warning

See section [5.3.1 Warnings and notes for electrical connection](#) for warnings and notes relating to electrical connection.

Connect the terminals of the servomotor according to the terminal connection diagram of the controller.

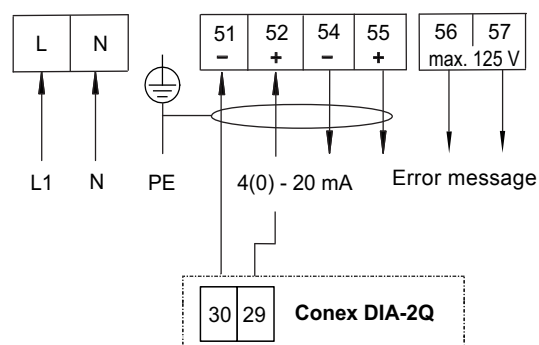


Fig. 15 Terminal connection diagram Conex® DIA-2Q

5.3.4 Connecting the external controller Conex® DIS-2Q



Warning

See section [5.3.1 Warnings and notes for electrical connection](#) for warnings and notes relating to electrical connection.

1. Unscrew the screws of the servomotor hood, and remove the hood.
2. Establish electrical connections according to the diagram shown in fig. 16.
 - Connect terminals 51 and 52 and, if required, terminals 54 and 55, in accordance with the diagram.
 - Connect terminals 56 and 57 in accordance with requirements; maximum voltage 125 V.
 - For low voltages (setpoint/actual value signal lines) use separate shielded lines with a minimum cross-section of 0.5 mm² and a maximum length of 1000 m.
 - The screen must be connected on one side to the enclosure earth (earth terminal).



Warning

The internal wiring of the servomotor must never be modified.

3. Close the hood again and tighten the hood screws.

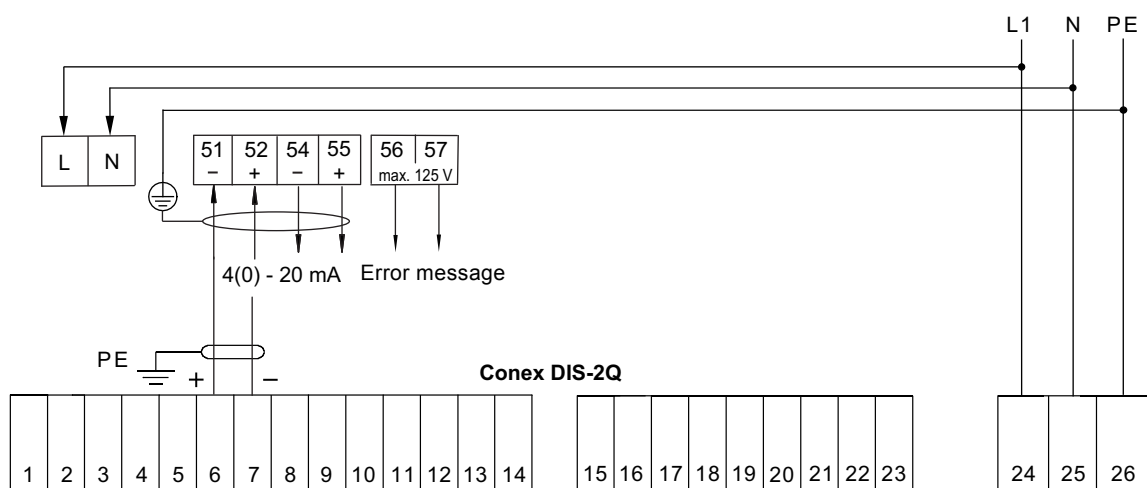


Fig. 16 Terminal connection diagram Conex® DIS-2Q

Pos.	Description
L	Phase
N	Neutral
PE	PE (earth)
51	Signal input (-)
52	Signal input (+)
54	Signal output (-)
55	Signal output (+)
56	Collective error message
57	

5.4 Connecting an external controller to a servomotor with reversing potentiometer

5.4.1 Terminal connection diagram servomotor with reversing potentiometer

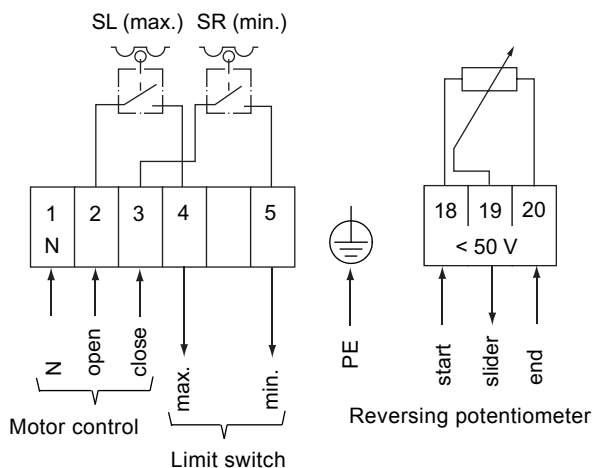


Fig. 17 Servomotor with reversing potentiometer

Pos.	Description
1 N	Neutral
2	Phase Supply voltage input (direction open/max.)
3	Phase Supply voltage input (direction close/min.)
4	Supply voltage output (final position max.)
5	Supply voltage output (final position min.)
PE	PE (earth)
18	Start
19	Slider
20	End

5.4.2 Connecting the external controller Conex® DIS-2Q



Warning

See section 5.3.1 [Warnings and notes for electrical connection](#) for warnings and notes relating to electrical connection.

1. Unscrew the screws of the servomotor hood, and remove the hood.
2. Establish electrical connections according to the diagram shown in fig. 18.
 - Connect terminals 18, 19 and 20 in accordance with requirements; voltage < 50 V.
3. Determine the direction of rotation and correct it, if necessary.

Determining the direction of rotation

According to the internal wiring, the direction of rotation (looking through the drive towards the output shaft) and limit switches are assigned as follows:

- If mains voltage is present at terminal 2, the output shaft rotates counter-clockwise (direction open/max.).
 - This direction of rotation is restricted by the upper switch (max.). If the switch is activated, the mains voltage is present at terminal 4.
- If mains voltage is present at terminal 3, the output shaft rotates clockwise (direction close/min.).
 - This direction of rotation is restricted by the lower switch (min.). If the switch is activated, the mains voltage is present at terminal 5.
- If the drive rotates in the opposite direction to the control commands, swap over the connections of terminals 2 and 3.



Warning

The internal wiring of the servomotor must never be modified.

4. Close the hood again and tighten the hood screws.

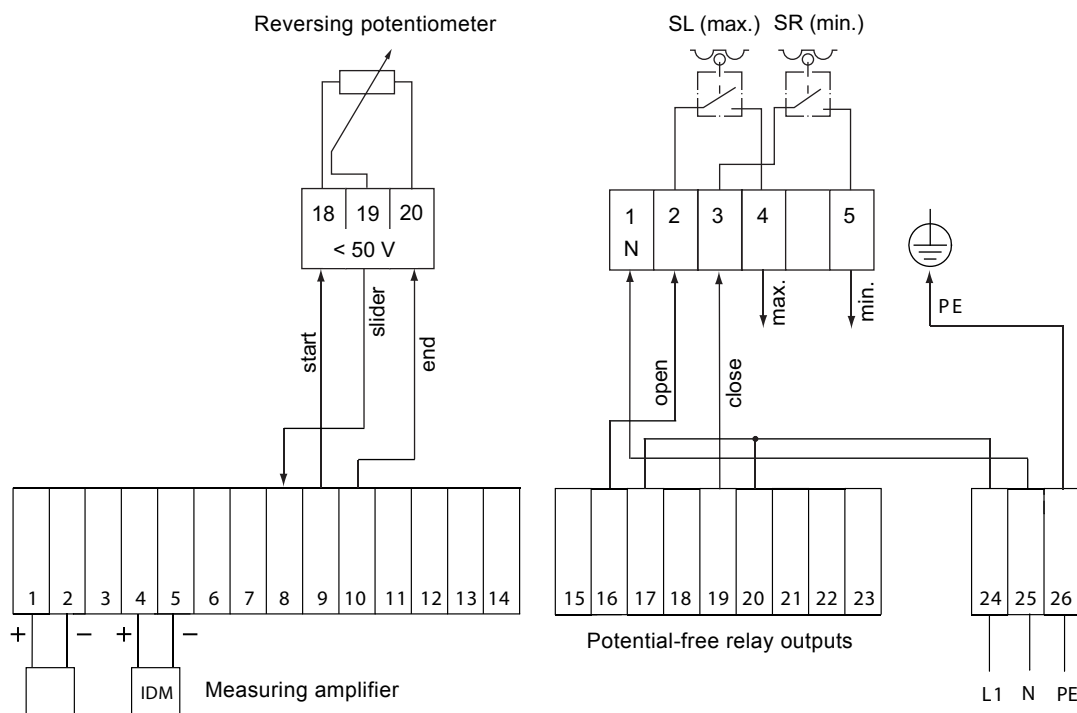


Fig. 18 Terminal connection diagram Conex® DIS-2Q

5.5 Installation example

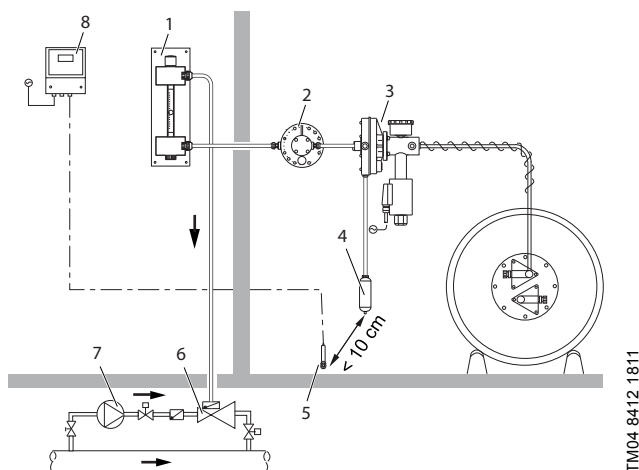


Fig. 19 Dosing system with one drum

Pos.	Description
1	Dosing regulator
2	Safety valve
3	Vacuum regulator with liquid trap
4	Chlorine gas adsorption filter
5	Gas sensor
6	Injector with non-return valve
7	Water apparatus
8	Gas warning device

6. Startup



Warning

The chlorine gas dosing system can only be started after its good condition has been checked by an expert, see [2.3 Qualification of the users](#), and especially gas-leading parts have been checked for their tightness.

Check the tightness of the entire system before start-up. Check the tightness of the valves using ammonia water (small plastic bottle). White mist indicates leakages.

6.1 Checking the tightness

Due to the heavy corrosiveness of humid chlorine gas, leakages increase quickly. Therefore, even the smallest leakage must be eliminated immediately.



Warning

Check the tightness only if the entire system is ready for start-up.

Danger of chlorine gas escape!

6.1.1 Checking the chlorine solution lines and the injector

Observe the installation and operating instructions of the injector.

6.1.2 Checking the vacuum lines

Vacuum lines are all lines between vacuum regulator and injector.

1. Close all chlorine container valves.
2. Close the rate valve.
3. Open the shut-off valve at the injection unit.
4. Open the motive water valve.
5. Switch on the booster pump.
6. Open the rate valve.
 - If the floater shows a gas flow, or the vacuummeter indicates more than -9 m water column, the vacuum lines are leaky.
7. Close the rate valve.
8. Switch off the booster pump.
9. Close the motive water valve.
10. Close the shut-off valve at the injection unit.
11. Check the vacuum lines and connections. Re-tighten them carefully, if necessary.
12. Repeat the tightness check.
 - If the floater shows no gas flow, and the vacuummeter indicates -9 m water column or less, the vacuum lines are tight.

Reasons for insufficient **operating vacuum**:

Note	Injector too small, defective, or blocked.
-------------	---

Booster pump too small, or defective.

6.1.3 Checking the gas pressure lines

Gas pressure lines are all lines leading from the chlorine containers to the vacuum regulator.

1. If the system is equipped with a nitrogen rinsing device, check the tightness roughly with nitrogen.
2. Detailed checking is made with ammonia.

Checking the tightness with nitrogen

1. Close all chlorine container valves.
2. Open container connection valves and all shut-off valves up to the chlorine gas dosing system.
3. Open the connection valve of the nitrogen cylinder.
4. Slowly open the valve of the nitrogen cylinder, until the lines have a pressure of about 10 bar (read at the pressure gauge of the vacuum regulator).



Warning

Maximum nitrogen pressure 16 bar.

Danger of damages and leakages when exceeded.

5. Apply soap water to all components under pressure.
 - If bubbles form and/or pressure drops at the pressure gauge, the pressure lines are leaky.
6. Depressurise the system.
7. Eliminate the leakage.
8. Repeat the tightness check.
9. If no bubbles form, and the pressure at the pressure gauge does not drop significantly within one hour, the pressure lines are tight.

Checking the tightness with ammonia



Warning

Ammonia is irritating to eyes, respiratory system and skin.

Ammonia is very toxic for water organisms.

When checking the tightness with ammonia, wear tight protective goggles, gloves and clothes.



TM04 0699 0908

Fig. 20 Hazard notes for ammonia

Even very small leakages become evident with ammonia vapour, which reacts with chlorine gas forming white mist.



Warning

Liquid ammonia must not come in contact with lines or parts of the system, because it is very corrosive.

Remove drops immediately with a dry cloth.

Checking the pressure connections (after change of container)

1. Open the container valve and immediately close it again.
2. Open the flexible plastic bottle filled with ammonia water.
3. Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up. Slowly pass the open ammonia bottle along gas-leading parts.
4. If white mist forms, the pressure connection is leaky.

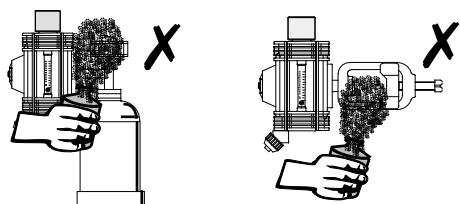


Fig. 21 Ammonia and chlorine gas form white mist

5. Depressurise the system.
6. Eliminate the leakage.
7. Repeat the tightness check.
8. If no white mist forms, the pressure connection is tight.

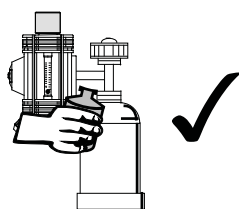


Fig. 22 Pressure connection is tight

TM04 0710 0908

TM04 0711 0908

Checking the inlet valve

1. Loosen the union nut at the overpressure connection of the vacuum regulator, and remove the overpressure line.
2. Close the inlet valve.
3. Open the chlorine container valve.
4. Open the flexible plastic bottle filled with ammonia water.
5. Slightly press the bottle in a pumping manner, allowing the ammonia mist to rise up. Slowly pass the open ammonia bottle along gas-leading parts.
6. If white mist forms, the inlet valve is leaky.
7. Depressurise the system.
8. Check the inlet valve and repair it.
9. Repeat the tightness check.
10. If no white mist forms, the inlet valve is tight.

6.2 Chlorine extraction

6.2.1 Connecting the chlorine container

Chlorine containers must be stored at least for 8 hours in the container room, so that the content can adapt to the ambient temperature.



Warning

Chlorine containers must not have a higher temperature than other parts of the system. Danger of liquefaction and escape of chlorine!

1. Turn chlorine drums on their support, until the dip pipe and the riser pipe are placed vertically (observe the markings on the drum).
2. Protect chlorine containers from turning over or rolling away.
3. Dry the piping and the extraction system with dry nitrogen or dry air.
4. No foreign matter must get into the system.
5. Provide the pressure connection with a new gasket.
6. Connect the chlorine container. When using drums, observe that the lower valve is normally intended for the extraction of liquid chlorine.
7. Slowly open the container valve.

6.2.2 Container valves

Operate the valves at the chlorine containers only manually without force. Close the valves of full or empty chlorine containers with closing nut (marked chlorine or Cl_2) and gasket.



Warning

Do not open container valves using force!

Stuck valves

Stuck valves can be loosened by wrapping a cloth with warm water around the valve. If this is not possible, return the container to the manufacturer.

- Never use an open flame to heat up the valve!
- Never use wrench lengthening!

6.2.3 Checking the tightness

Before start-up, check the tightness of the entire system, see section [6.1 Checking the tightness](#).

6.2.4 Chlorine extraction

Do not completely evacuate the chlorine container.
A residual pressure of approx. 2 bar must remain in the container for the following reasons:

- Caution** To avoid penetration of humidity, which will cause corrosion.
- To avoid withdrawal of chlorine gas pollutants, which will cause clogging of the pressure reducing valve in the long run.

Chlorine gas extraction

At 15 °C about 1 % of the content of a container can be extracted per hour.

- Caution** When extracting more, there is danger of formation of ice on containers and piping, which will cause malfunction.

The table shows the maximum extraction quantity depending on the used container size.

Container	Container size [kg]	Extraction quantity [g/h]
Cylinder	50	max. 500
	65	max. 650
Drum	500	max. 5000
	1000	max. 10000

In case of higher chlorine requirement, several chlorine containers of the same temperature have to be connected with header lines.

Extraction of liquid chlorine

With chlorine gas dosing systems, the extraction of liquid chlorine is only possible from chlorine drums. The valve on the dip pipe of the drum must be connected, and an evaporator must be used.

6.2.5 After chlorine extraction

- Caution** In order to avoid the penetration of humidity causing corrosion, close the connecting lines directly after chlorine extraction.

1. If the container is emptied down to a residual pressure of 2 bar, the container valve must be closed, in order to avoid the withdrawal of pollutants or penetration of humidity.
2. Disconnect the container from the system.
3. Mount the protective nut on the valve.
4. Mount the protective cap.

7. Operation

The dosing flow is adjusted at the rate valve of the dosing regulator. In versions with optional servomotor the dosing flow can also be adjusted via an external controller with servomotor.

7.1 Switching on the chlorine gas dosing system

1. Close the rate valve at the dosing regulator.
2. Open the shut-off valve at the injection unit.
3. Open the motive water valves.
4. Open the motive water supply of the injector.
5. Open the chlorine container valve.
6. Slowly open the rate valve of the dosing regulator using the adjustment knob (1), until the ball in the measuring glass indicates the desired gas flow.

If the dosing flow is adjusted via an external controller, no adjustment can be made at the dosing regulator.

Note

See also the manual of the external controller.

7.2 Setting the dosing flow

To increase the dosing flow, turn the adjustment knob very slowly counter-clockwise.

- Caution** If the maximum dosing flow is reached, don't turn the adjustment knob any further, because there is no locking.

To reduce the dosing flow, turn the adjustment knob very slowly clockwise.

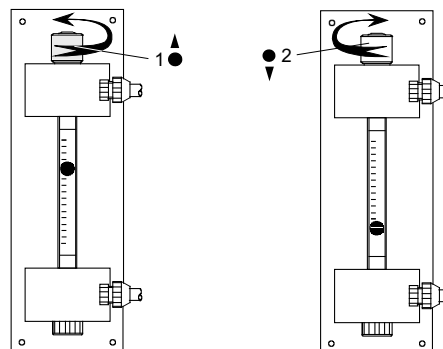


Fig. 23 Setting the dosing flow

Pos.	Description
1	Turning the adjustment knob counter-clockwise will increase the dosing flow.
2	Turning the adjustment knob clockwise will reduce the dosing flow.

7.3 Reading the dosing flow

The dosing flow is read at the top of the ball in the measuring glass.

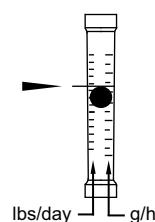


Fig. 24 Reading the dosing flow

The scale of the measuring glass is adjusted to a gas temperature of 20 °C.

Note

In case of extremely different temperatures the dosing flow may differ from the value being shown.

7.4 Servomotor

7.4.1 Connecting the servomotor to the rate valve

1. Move back the retention pin (2).
2. The coupling (1) slides down.
3. Turn the adjustment knob (3), until the coupling snaps in.
The servomotor is connected.

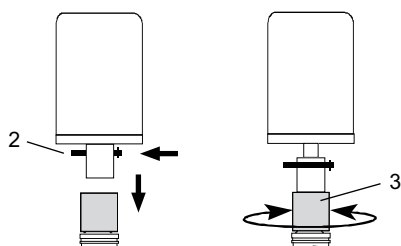


Fig. 25 Connecting the servomotor

7.4.2 Disconnecting the servomotor from the rate valve

1. Switch off the automatic control.
2. Push the coupling (1) upwards.
3. Move the retention pin (2) through the coupling, until it cannot move down any longer.
4. The servomotor is disconnected from the rate valve.
5. Now the dosing flow can be adjusted manually.

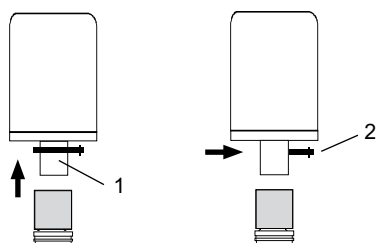


Fig. 26 Disconnecting the servomotor

7.5 Switching off the chlorine gas dosing system

7.5.1 Emergency stop in case of gas escape



Warning

If gas escapes, immediately leave the room and put on protective equipment.

Start counter measures according to local safety regulations.

7.5.2 Emergency stop in case of small leakages

1. Put on protective equipment.
2. Immediately close the chlorine container valve.
3. Let the system run until all parts are evacuated.
4. Switch off the system as described in the following.



Warning

Repairs of components of the system are to be carried out only by authorised persons.

7.5.3 Switching off while the system is running

1. Close the chlorine container valve.
2. Let the system run, until the measuring glass of the dosing regulator shows no more gas flow.
3. Close the rate valve at the dosing regulator.
4. Turn off the motive water supply.
5. Close the motive water valves.
6. Close the shut-off valves before and after the injector.

7.6 Changing the chlorine container while the system is running

The red mark (empty indication) at the vacuum regulator shows that the minimum pressure of the chlorine container is reached. The container must be replaced.



Warning

Put on the required protective equipment.

1. Close the chlorine container valve.
2. Let the system run, until the measuring glass of the dosing regulator shows no more gas flow.
3. Close the rate valve at the dosing regulator.
4. Turn off the motive water supply.
5. Close the motive water valves.
6. Close the shut-off valves before and after the injector.
7. Unscrew the vacuum regulator from the connection of the container.
– Place the regulator into the wall support, if existing.
8. Screw the protective cap onto the empty chlorine container; put the empty containers away, and protect them from rolling away or turning over.
9. Position the new chlorine container, and fix it correctly.
10. Remove the protective cap from the new chlorine container.
– All connections must be absolutely clean and dry.



Warning

Replace the gaskets at any change of chlorine container.

Never use the vacuum regulator without filter!

11. Insert the gasket at the pressure connection of the vacuum regulator.
– Replace the filter, if necessary.
12. Screw the vacuum regulator onto the chlorine container.

Caution

Do not align the regulator by turning the housing!
If necessary, hold up with a 13-mm fork wrench.



Warning

Check the tightness before re-starting the system.

13. Check the tightness as described in section [6.1 Checking the tightness](#).
14. Re-start the system.

8. Maintenance

Intervals for cleaning and maintenance

- At least every 12 months.
- Before every start-up.
- In case of malfunction.



Warning

Maintenance has to be carried out only by authorised and qualified personnel.

Before starting cleaning and maintenance work, switch off the entire system.

Danger of chlorine gas escape!



Warning

Make sure that the system cannot be started accidentally while working at it.



Warning

Before re-starting, check the tightness.
Danger of chlorine gas escape!

9. Fault finding

Fault	Cause	Remedy
1. Maximum dosing flow is not reached.	a) Insufficient injector vacuum.	Check the injector. Repair it, if necessary. Observe the manual of the injector.
		Check the motive water pump. Repair it, if necessary.
		Check the dirt trap before the injector. Clean it, if necessary.
	b) Leakage in the vacuum line between dosing regulator and injector.	Eliminate the leakage.
	c) The vacuum line between dosing regulator and injector is too long.	For details on vacuum lines, see section 5.1.2 Vacuum lines between dosing regulator and injector
	d) Soiled vacuum line.	Replace the vacuum line.
	e) Servomotor does not reach maximum dosing capacity.	Readjust the servomotor and/or the external controller.
	f) Soiled filter at the pressure connection of the dosing regulator.	Call the service.
	g) Closed chlorine container valve.	Open the chlorine container valve.
2. Leaky pressure connection.	h) Empty chlorine container.	Replace empty chlorine container by a full one.
	i) Adjusting spindle of the dosing regulator not suitable (diameter too small).	Call the service.
	a) Gasket at the pressure connection damaged.	Replace the gasket.
	b) Sealing surfaces soiled with residues of old gaskets.	Clean the sealing surfaces.
	c) Sealing surface at the chlorine container damaged.	Use another chlorine container.
3. Gas escapes at the overpressure line.	d) Pressure connection loose.	Tighten the pressure connection.
	e) Sealing surface at the pressure connection of the vacuum regulator damaged.	Call the service.
4. Despite correctly displayed dosing capacity, the desired chlorine solution concentration is not reached.	a) Re-liquefied gas gets into the inlet valve.	Reduce the extraction amount, ensure a correct temperature sequence.
	b) Inlet valve soiled or damaged.	Call the service.
5. The ball in the measuring glass is caught.	a) Measuring glass or ball soiled.	Call the service.
6. Formation of ice on the chlorine container.	a) Extraction quantity is set too high. If more than 1 % of the content is extracted per hour, there is danger of formation of ice at the container.	Use a dosing system with several chlorine containers. Call the service.
	b) Chlorine gas dosing system too small.	Use a bigger chlorine gas dosing system.
7. Water in the measuring glass.	a) Diaphragm non-return valve is defective.	Call the service.
8. Servomotor does not run.	a) Servomotor is switched to manual operation.	Switch servomotor to automatic operation.
9. Servomotor does not regulate within the entire capacity range.	a) Servomotor and/or controller adjusted incorrectly.	Call the service.
10. Servomotor rotates in the wrong direction.		
11. Vacuum varies despite constant operating conditions.	a) Spring in the differential pressure regulator soiled or damaged.	
12. The dosing capacity varies despite constant operating conditions.	a) Diaphragm in the differential pressure regulator soiled or damaged.	

10. Disposal

This product and all its associated parts must be disposed of in an environmentally friendly manner. Use appropriate waste collection services. If this is not possible, contact the nearest Grundfos company or service workshop.

Declaration of conformity

GB: EU declaration of conformity

We, Grundfos, declare under our sole responsibility that the products VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, to which the declaration below relates, are in conformity with the Council Directives listed below on the approximation of the laws of the EU member states.

DE: EU-Konformitätserklärung

Wir, Grundfos, erklären in alleiniger Verantwortung, dass die Produkte VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, auf die sich diese Erklärung beziehen, mit den folgenden Richtlinien des Rates zur Angleichung der Rechtsvorschriften der EU-Mitgliedsstaaten übereinstimmen.

FR: Déclaration de conformité UE

Nous, Grundfos, déclarons sous notre seule responsabilité, que les produits VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, auxquels se réfère cette déclaration, sont conformes aux Directives du Conseil concernant le rapprochement des législations des États membres UE relatives aux normes énoncées ci-dessous.

PT: Declaração de conformidade UE

A Grundfos declara sob sua única responsabilidade que os produtos VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, aos quais diz respeito a declaração abaixo, estão em conformidade com as Directivas do Conselho sobre a aproximação das legislações dos Estados Membros da UE.

RS: Deklaracija o usklađenosti EU

Mi, kompanija Grundfos, izjavljujemo pod punom vlastitom odgovornošću da je proizvod VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, na koji se odnosi deklaracija ispod, u skladu sa dole prikazanim direktivama Saveta za usklađivanje zakona država članica EU.

TR: AB uygunluk bildirgesi

Grundfos olarak, aşağıdaki bildirim konusu olan VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 ürünlerinin, AB Üye ülkelerinin direktiflerinin yakınlaştırılmasıyla ilgili durumun aşağıdaki Konsey Direktifleriyle uyumlu olduğunu ve bununla ilgili olarak tüm sorumluluğun bize ait olduğunu beyan ederiz.

CZ: Prohlášení o shodě EU

My firma Grundfos prohlašujeme na svou plnou odpovědnost, že výrobky VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, na které se toto prohlášení vztahuje, jsou v souladu s níže uvedenými ustanoveními směrnice Rady pro sblížení právních předpisů členských států Evropského společenství.

ES: Declaración de conformidad de la UE

Grundfos declara, bajo su exclusiva responsabilidad, que los productos VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148 a los que hace referencia la siguiente declaración cumplen lo establecido por las siguientes Directivas del Consejo sobre la aproximación de las legislaciones de los Estados miembros de la UE.

PL: Deklaracja zgodności UE

My, Grundfos, oświadczamy z pełną odpowiedzialnością, że nasze produkty VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, których deklaracja niniejsza dotyczy, są zgodne z następującymi dyrektywami Rady w sprawie zbliżenia przepisów prawnych państw członkowskich.

RO: Declarația de conformitate UE

Noi Grundfos declarăm pe propria răspundere că produsele VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, la care se referă această declarație, sunt în conformitate cu Directivele de Consiliu specificate mai jos privind armonizarea legilor statelor membre UE.

RU: Декларация о соответствии нормам ЕС

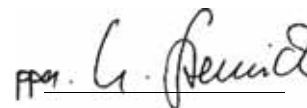
Мы, компания Grundfos, со всей ответственностью заявляем, что изделия VGA-111, VGA-113, VGA-117, VGA-146, VGB-103, VGS-141, VGS-143, VGS-145, VGS-147, VGS-148, к которым относится нижеприведенная декларация, соответствуют нижеприведенным Директивам Совета Евросоюза о тождественности законов стран-членов ЕС.

- Machinery Directive (2006/42/EC). Standards used: DIN EN ISO 12100:2010.
- Low Voltage Directive (2014/35/EU). Standards used: EN 60204-1:2007
EN 61010-1:2011-07*
- EMC Directive (2014/30/EU)*. Standards used: EN 55014-1:2006+A1:2009+A2:2011
EN 55014-2:2016
EN 61000-6-1:2007
EN 61000-6-3:2011
- Other regulations applied: DIN 19606:2010-09
- RoHS Directives (2011/65/EU and 2015/863/EU). Standard used: EN 50581:2012

* Only valid for products with electrical components

This EU declaration of conformity is only valid when published as part of the Grundfos installation and operating instructions (publication numbers 95714202, 95714224, 98028175, 95714246, 95714278, 95713929, 95714262, 95714296).

Pfinztal, 1st March 2018



Ulrich Sternick
Technical Director
Grundfos Water Treatment GmbH
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